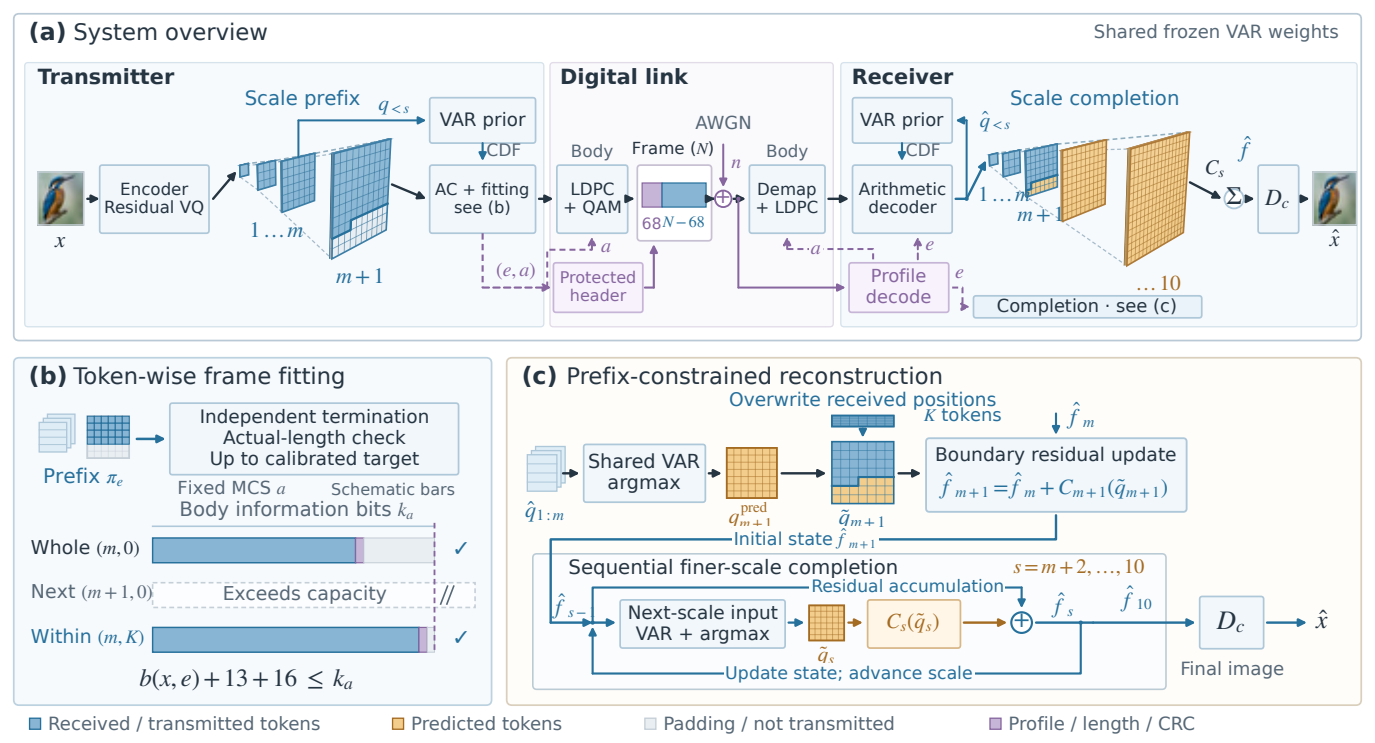

(a) System overview
Shared frozen VAR weights
Transmitter
Scale prefix
$q_{<s}$
VAR prior
CDF
Encoder Residual VQ
AC + fitting see (b)
x
$1 \dots m$
$m+1$
(e, a)
Digital link
AWGN
Body
Frame (N)
Body
n
LDPC + QAM
$68N-68$
Demap + LDPC
a
Protected header
Receiver
VAR prior
CDF
Arithmetic decoder
Scale completion
$\hat{q}_{<s}$
$1 \dots m$
$m+1$
$\dots 10$
$\hat{f}$
C_s
Σ
D_c
$\hat{x}$
e
Profile decode
Completion · see (c)
(b) Token-wise frame fitting
Prefix π_e
Independent termination
Actual-length check
Up to calibrated target
Fixed MCS a
Schematic bars
Body information bits k_a
Whole $(m, 0)$
Next $(m+1, 0)$
Exceeds capacity
Within (m, K)
$b(x, e) + 13 + 16 \le k_a$
(c) Prefix-constrained reconstruction
Overwrite received positions
K tokens
$\hat{f}_m$
$\hat{q}_{1:m}$
Shared VAR argmax
q^{pred}_{m+1}
$\tilde{q}_{m+1}$
Boundary residual update
$\hat{f}_{m+1} = \hat{f}_m + C_{m+1}(\tilde{q}_{m+1})$
Initial state $\hat{f}_{m+1}$
Sequential finer-scale completion
$s = m+2, \dots, 10$
Residual accumulation
$\hat{f}_{s-}$
Next-scale input VAR + argmax
$\tilde{q}_s$
$C_s(\tilde{q}_s)$
$\hat{f}_s$
$\hat{f}_{10}$
D_c
$\hat{x}$
Final image
Update state; advance scale
Received / transmitted tokens
Predicted tokens
Padding / not transmitted
Profile / length / CRC